

1. Let the eccentricity of an ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  is reciprocal to that of the hyperbola  $2x^2 - 2y^2 = 1$ . If the ellipse intersects the hyperbola at right angles, then square of length of the latus-rectum of the ellipse is \_\_\_\_\_.  
[2023 (06 Apr Shift 2)]
2. Let the ellipse  $E : x^2 + 9y^2 = 9$  intersect the positive  $x$ - and  $y$ -axes at the points  $A$  and  $B$  respectively. Let the major axis of  $E$  be a diameter of the circle  $C$ . Let the line passing through  $A$  and  $B$  meet the circle  $C$  at the point  $P$ . If the area of the triangle with vertices  $A$ ,  $P$  and the origin  $O$  is  $\frac{m}{n}$ , where  $m$  and  $n$  are coprime, then  $m - n$  is equal to \_\_\_\_\_.  
[2023 (10 Apr Shift 1)]  
(1) 16  
(2) 15  
(3) 17  
(4) 18
3. Let a circle of radius 4 be concentric to the ellipse  $15x^2 + 19y^2 = 285$ . Then the common tangents are inclined to the minor axis of the ellipse at the angle \_\_\_\_\_.  
[2023 (10 Apr Shift 2)]  
(1)  $\frac{\pi}{3}$   
(2)  $\frac{\pi}{4}$   
(3)  $\frac{\pi}{6}$   
(4)  $\frac{\pi}{12}$
4. Consider ellipses  $E_k : kx^2 + ky^2 = 1, k = 1, 2, \dots, 20$ . Let  $C_k$  be the circle which touches the four chords joining the end points (one on minor axis and another on major axis) of the ellipse  $E_k$ . If  $r_k$  is the radius of the circle  $C_k$ , then the value of  $\sum_{k=1}^{20} \frac{1}{r_k^2}$  is \_\_\_\_\_.  
[2023 (11 Apr Shift 1)]  
(1) 3080  
(2) 2870  
(3) 3210  
(4) 3320
5. Let  $P\left(\frac{2\sqrt{3}}{\sqrt{7}}, \frac{6}{\sqrt{7}}\right)$ ,  $Q$ ,  $R$  and  $S$  be four points on the ellipse  $9x^2 + 4y^2 = 36$ . Let  $PQ$  and  $RS$  be mutually perpendicular and pass through the origin. If  $\frac{1}{(PQ)^2} + \frac{1}{(RS)^2} = \frac{p}{q}$ , where  $p$  and  $q$  are coprime, then  $p + q$  is equal to \_\_\_\_\_.  
[2023 (12 Apr Shift 1)]  
(1) 147  
(2) 143  
(3) 137  
(4) 157
6. Let an ellipse with centre  $(1, 0)$  and latus rectum of length  $\frac{1}{2}$  have its major axis along  $x$ -axis. If its minor axis subtends an angle  $60^\circ$  at the foci, then the square of the sum of the lengths of its minor and major axes is equal to \_\_\_\_\_.  
[2023 (15 Apr Shift 1)]

## ANSWER KEYS

1. (2)      2. (3)      3. (1)      4. (1)      5. (4)      6. (9)

1. (2)

Given,

The eccentricity of an ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  is reciprocal to that of the hyperbola  $2x^2 - 2y^2 = 1$ ,

Now eccentricity of rectangular hyperbola  $2x^2 - 2y^2 = 1$  is  $e_H = \sqrt{2}$

So,  $e_e = \frac{1}{\sqrt{2}}$

Now, focus of hyperbola  $= (\pm 1, 0)$

Now given that both curve intersect orthogonally, so ellipse and hyperbola are confocal

So, for ellipse  $ae_e = 1 \Rightarrow a = \sqrt{2}$

Now length of latusrectum L. R.  $= \frac{2b^2}{a} = 2a(1 - e_e^2)$

$= 2\sqrt{2} \cdot \frac{1}{2} = \sqrt{2}$

2. (3)

Given,

Ellipse  $E : x^2 + 9y^2 = 9 \dots\dots\dots(i)$

Now point  $A(3, 0)$  &  $B(0, 1)$  which is the intersection of given ellipse with positive axis,

So, equation of line passing through  $A$  &  $B$  is given by,

$L : \frac{x}{3} + \frac{y}{1} = 1 \Rightarrow x = 3 - 3y \dots\dots\dots(ii)$

Now equation of Circle with diametric point  $(-3, 0)$  &  $(3, 0)$  is given by,

$C : x^2 + y^2 = 9 \dots\dots\dots(iii)$

Let  $Q$  be foot of perpendicular from  $P$  upon major axis,

So, from (ii) & (iii) we get,

$(3 - 3y)^2 + y^2 = 9$

$\Rightarrow y = \frac{9}{5}, 0$

Hence,  $PQ = \frac{9}{5}$

Now Area of triangle will be,

$= \frac{1}{2} \times OA \times PQ = \frac{1}{2} \times 3 \times \frac{9}{5} = \frac{27}{10}$

Hence,  $m - n = 17$

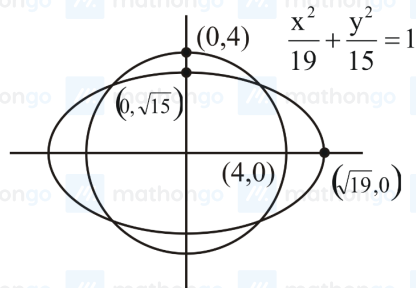
3. (1)

Given,

$$15x^2 + 19y^2 = 285$$

$$\Rightarrow \frac{x^2}{19} + \frac{y^2}{15} = 1$$

Now plotting the diagram with given circle of radius 4 we get,



Now, let the equation of tangent of ellipse be  $y = mx \pm \sqrt{19m^2 + 15}$  {as  $c^2 = a^2m^2 + b^2$ }

If it is tangent to circle  $x^2 + y^2 = 16$  then using the formula,  $c^2 = r^2(1 + m^2)$  we get,

$$\frac{\sqrt{19m^2 + 15}}{\sqrt{1 + m^2}} = 4$$

$$\Rightarrow 19m^2 + 15 = 16 + 16m^2$$

$$\Rightarrow 3m^2 = 1$$

$$\Rightarrow m = \pm \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta = 30^\circ \text{ with } x\text{-axis,}$$

Hence, Angle made by tangent with minor axis i.e., with  $y$  axis is  $\frac{\pi}{3}$

4. (1)

Given,

$$E_k : kx^2 + ky^2 = 1$$

$$\Rightarrow E_k : \frac{x^2}{\left(\frac{1}{\sqrt{k}}\right)^2} + \frac{y^2}{\left(\frac{1}{k}\right)^2} = 1$$

Now equation of the chord joining the points  $\left(\frac{1}{\sqrt{k}}, 0\right)$  &  $\left(0, \frac{1}{k}\right)$  will be,

$$L_k : \frac{x}{\left(\frac{1}{\sqrt{k}}\right)} + \frac{y}{\left(\frac{1}{k}\right)} = 1$$

$$\Rightarrow \sqrt{k}x + ky - 1 = 0$$

Now  $r_k$  = Perpendicular distance of  $L_k$  from  $(0, 0)$  we get,

$$r_k = \left| \frac{-1}{\sqrt{k+k^2}} \right|$$

$$\Rightarrow r_k^2 = \frac{1}{k+k^2}$$

Now putting the value of  $r_k^2$  in  $\sum_{k=1}^{20} \frac{1}{r_k^2}$  we get,

$$\sum_{k=1}^{20} \frac{1}{r_k^2} = \sum_{k=1}^{20} k + k^2 = \frac{20 \times 21}{2} + \frac{20 \times 21 \times 41}{6}$$

$$= 210 + 2870 = 3080$$

5. (4)

Given,

$P\left(\frac{2\sqrt{3}}{\sqrt{7}}, \frac{6}{\sqrt{7}}\right)$ ,  $Q$ ,  $R$  and  $S$  be four points on the ellipse  $\frac{x^2}{4} + \frac{y^2}{9} = 1$

Now,  $OP = r_1 = \sqrt{\left(\frac{2\sqrt{3}}{\sqrt{7}}\right)^2 + \left(\frac{6}{\sqrt{7}}\right)^2} = \sqrt{\frac{48}{7}}$  where  $O$  is origin,

Let  $P$  be  $(r_1 \cos \theta, r_1 \sin \theta)$

$P$  lies on ellipse, so we get,

$$\frac{r_1^2 \cos^2 \theta}{4} + \frac{r_1^2 \sin^2 \theta}{9} = 1$$

$$\Rightarrow \frac{\cos^2 \theta}{4} + \frac{\sin^2 \theta}{9} = \frac{7}{48} \dots (i)$$

Let  $R$  be  $(-r_2 \sin \theta, r_2 \cos \theta)$  as  $PQ$  &  $RS$  are perpendicular and pass through origin,

$$\text{So, } \frac{r_2^2 \sin^2 \theta}{4} + \frac{r_2^2 \cos^2 \theta}{9} = 1$$

$$\Rightarrow \frac{\sin^2 \theta}{4} + \frac{\cos^2 \theta}{9} = \frac{1}{r_2^2} \dots (ii)$$

Now adding equation (i) & (ii) we get,

$$\frac{1}{r_2^2} = \frac{1}{4} + \frac{1}{9} - \frac{7}{48} = \frac{31}{144}$$

Now solving,

$$\frac{1}{PQ^2} + \frac{1}{RS^2} = \frac{1}{4} \left( \frac{1}{OP^2} + \frac{1}{OR^2} \right)$$

$$\Rightarrow \frac{1}{PQ^2} + \frac{1}{RS^2} = \frac{1}{4} \left( \frac{1}{r_1^2} + \frac{1}{r_2^2} \right)$$

$$\Rightarrow \frac{1}{PQ^2} + \frac{1}{RS^2} = \frac{1}{4} \left( \frac{7}{48} + \frac{31}{144} \right) = \frac{13}{144} = \frac{p}{m}$$

$$\therefore p + m = 157$$

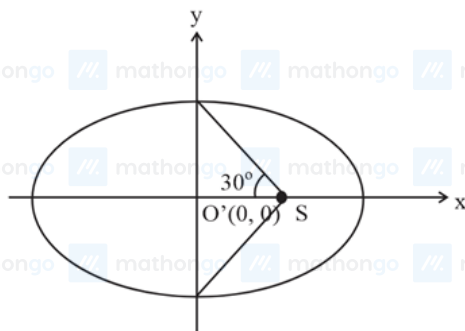
6. (9)

Given,

An ellipse with centre  $(1, 0)$  and latus rectum of length  $\frac{1}{2}$  have its major axis along x-axis

And its minor axis subtends an angle  $60^\circ$  at the foci,

So, on plotting the diagram we get,



Now from diagram we get,

$$\frac{2b^2}{a} = \frac{1}{2} \Rightarrow b^2 = \frac{a}{4} \dots (1)$$

$$\text{And } \tan 30^\circ = \frac{b}{ae}$$

$$\Rightarrow \left( \frac{1}{\sqrt{3}} \right)^2 = \left( \frac{b}{ae} \right)^2$$

$$\Rightarrow \frac{1}{3} = \frac{b^2}{a^2 - b^2}$$

$$\Rightarrow a^2 - b^2 = 3b^2$$

$$\Rightarrow b^2 = \frac{a^2}{4} \dots (2)$$

Now from equation (1) & (2) we get,

$$\Rightarrow a = 1, b^2 = \frac{1}{4} \Rightarrow b = \frac{1}{2}$$

$$\text{Hence, } (2a + 2b)^2 = 9$$